## Getting familiar with basic SQL Syntax (DDL, DML Statements)

### 1. Create following table which has following structure:

**a. TABLE NAME: EMP**

| Field Name | Data Type | Null |
|---|---|---|
| EMPNO | NUMBER(4) PK | |
| ENAME | VARCHAR2(20) | |
| JOB | VARCHAR2(20) | |
| MGR | NUMBER(4) | YES |
| HIREDATE | DATE | |
| SAL | NUMBER(7,2) | |
| COMM | NUMBER(7,2) | YES |
| DEPTNO | NUMBER(2) FK | |

**TABLE NAME : DEPT**

| Field Name | Data Type | Null |
|---|---|---|
| DEPTNO PK | NUMBER(2) | |
| DNAME | VARCHAR2(20) | |
| LOC | VARCHAR2(20) | |

**b. TABLE NAME : SALGRADE**

| Field Name | Data Type | Null |
|---|---|---|
| GRADE | NUMBER(2) | |
| LOSAL | NUMBER (4) | |
| HISAL | NUMBER (4) | |

### 2. Insert following records in respective tables.

**Table Name: EMP**

| EMPNO | ENAME | JOB | MGR | HIREDATE | SAL | COMM | DEPTNO |
|---|---|---|---|---|---|---|---|
| 7369 | SMITH | CLERK | 7902 | 17-DEC-80 | 800 | | 20 |
| 7499 | ALLEN | SALESMAN | 7698 | 20-FEB-81 | 1600 | 300 | 30 |
| 7521 | WARD | SALESMAN | 7698 | 22-FEB-81 | 1250 | 500 | 30 |
| 7566 | JONES | MANAGER | 7839 | 02-APR-81 | 2975 | | 20 |
| 7654 | MARTIN | SALESMAN | 7698 | 28-SEP-81 | 1250 | 1400 | 30 |
| 7698 | BLAKE | MANAGER | 7839 | 01-MAY-81 | 2850 | | 30 |
| 7782 | CLARK | MANAGER | 7839 | 09-JUN-81 | 2450 | | 10 |
| 7788 | SCOTT | ANALYST | 7566 | 09-DEC-81 | 3000 | | 20 |
| 7839 | KING | PRESIDENT | | 17-NOV-81 | 5000 | | 10 |
| 7844 | TURNER | SALESMAN | 7698 | 08-SEP-81 | 1500 | 0 | 30 |
| 7876 | ADAMS | CLERK | 7788 | 12-JAN-83 | 1100 | | 20 |
| 7900 | JAMES | CLERK | 7698 | 03-DEC-81 | 950 | | 30 |
| 7802 | FORD | ANALYST | 7566 | 03-DEC-81 | 3000 | | 20 |
| 7934 | MILLER | CLERK | 7782 | 23-JAN-82 | 1300 | | 10 |

**Table Name: DEPT**

| DEPTNO | DNAME | LOC |
|---|---|---|
| 10 | ACCOUNTING | NEW YORK |
| 20 | RESEARCH | DALLA |
| 30 | SALES | CHICAGO |
| 40 | OPERATIONS | BOSTON |

**Table Name: SALGRADE**

| GRADE | HISAL | LOSAL |
|---|---|---|
| 1 | 700 | 1200 |
| 2 | 1201 | 1400 |
| 3 | 1401 | 2000 |
| 4 | 2001 | 3000 |
| 5 | 3001 | 9999 |

## 3. Solve the following queries.

| | |
|---|---|
| 1. | Find out the name of all employees. |
| 2. | Display all information of employee. |
| 3. | Display all information of employee whose job type is “SALESMAN”. |
| 4. | Display all information of employee whose Dept No is 20. |
| 5. | Display employee information whose hire date is in month of DEC. |
| 6. | Display employee information whose salary is greater than 3000. |
| 7. | Find the names of all employees having ‘A’ as the second letter in their names. |
| 8. | Display department information. |
| 9. | Display Salary Grade Information. |
| 10. | Display Name, Job of employee whose manager name is “BLAKE”. |
| 11. | Display Name, Salary of employee whose commission is null. |
| 12. | Display Name, Salary, Hiredate of employee whose Hiredate is in the year of 81. |
| 13. | Update salary by 5% whose department no is 10. |
| 14. | Update commission comm=sal*5% whose job type is “MANAGER”. |
| 15. | Display employee name whose name start with ‘A’. |
| 16. | Display employee details that have maximum salary. |
| 17. | Display employee details that have minimum salary. |
| 18. | Display average salary department wise. |
| 19. | Display employee details whose salary is between 1000 to 2000. |
| 20. | Display employee information department wise. |

| | |
|---|---|
| 21. | Display your name in upper, lower and title case. |
| 22. | Display all Employee names and their length after trimming them |
| 23. | Extract first 3 characters from your name and display the same. |
| 24. | Extract last 4 characters from your name and display the same. |
| 25. | Extract only 3 to 7 characters of your full name. |
| 26. | Find out the absolute values of the following: -20, 30, 25, -400, -13, -11, 34, -99. |
| 27. | Update the employee table so each employee name will contain XXXXX at the end of the same. |
| 28. | Now display the same such that they won't display XXXXX at the end of each name. |
| 29. | Round the following numbers by 5, 2, 1, 3, 4 decimal places: 4.091978,232.837733, 822.2325443. |

## Getting familiar with basic SQL Syntax (DDL, DML Statements)

**2. Create following table which has following structure:**

**a. TABLE NAME: CLIENT_MASTER**

| Field Name | Data Type | Constraint |
|---|---|---|
| CLIENT_NO | VARCHAR2(6) | Primary Key /First letter must Start with 'C'. |
| NAME | VARCHAR2(20) | Not null |
| ADDRESS1 | VARCHAR2(30) | |
| ADDRESS2 | VARCHAR2 (30) | |
| CITY | VARCHAR2(15) | |
| STATE | VARCHAR2 (15) | |
| PINCODE | NUMBER(6) | |
| BAL_DUE | NUMBER(10,2) | |

**a. TABLE NAME : PRODUCT_MASTER**

| Field Name | Data Type | Constraint |
|---|---|---|
| PRODUCT_NO | varchar2(6) | Primary key<br>Must start with P |
| DESCRIPTION | VARCHAR2(14) | NOT NULL |
| PROFIT_PERCENT | NUMBER(4,2) | NOT NULL |
| UNIT_MEASURE | VARCHAR2(10) | NOT NULL |
| QTY_ON_HAND | NUMBER(8) | NOT NULL |
| REORDER_LVL | NUMBER(8) | NOT NULL |
| SELL_PRICE | NUMBER(8,2) | NOT NULL,<br>Cannot be 0 |
| COST_PRICE | NUMBER(8,2) | NOT NULL,<br>Cannot be 0 |

### a. TABLE NAME : SALESMAN_MASTER

| Field Name | Data Type | Null |
|---|---|---|
| SALESMAN_NO | VARCHAR2(6) | Primary Key / first letter must start With 'S' |
| SALESMAN_NAME | VARCHAR2(20) | NOT NULL |
| ADDRESS1 | VARCHAR2(30) | NOT NULL |
| ADDRESS2 | VARCHAR2(30) | |
| CITY | VARCHAR2(20) | |
| PINCODE | VARCHAR2(6) | |
| STATE | VARCHAR2(20) | |
| SAL_AMT | NUMBER(8,2) | Not Null,cannot be 0 |
| TGT_TO_GET | NUMBER(6,2) | Not Null,cannot be 0 |
| YTD_SALES | NUMBER(6,2) | NOT NULL |
| REMARKS | VARCHAR2(60) | |

**Table Name :**sales_order

| Field Name | Data Type | Null |
|---|---|---|
| S_ORDER_NO | VARCHAR2(6) | Primary Key / first letter must start With 'O' |
| S_ORDER_DATE | DATE | |
| CLIENT_NO | VARCHAR2(6) | Foreign Key references client_no Of client_master table |
| DELY_ADDR | VARCHAR2(25) | |
| SALESMAN_NO | VARCHAR2(6) | Foreign Key references salesman_no Of salesman_master table. |
| DELY_TYPE | CHAR(1) | delivery : part(P)/full(F),Default 'F' |
| BILLED_YN | CHAR(1) | |
| DELY_DATE | DATE | cannot be less than s_order_date |
| ORDER_STATUS | VARCHAR2(10) | values ('In Process','Fulfilled', 'BackOrder','Canceled') |

**Perform Following Changes on Product_Master table**

1) Change Datatype of Product_no field from number(2) to varchar2(6).
2) Assign primary key to product_no field
3) Add Constraint to product_no field (Must start with P)

**Table Name :** sales_order_details

| Field Name | Data Type | Null |
|---|---|---|
| S_ORDER_NO | VARCHAR2(6) | Foreign Key references S_order_no of sales_order table. |
| PRODUCT_NO | VARCHAR2(6) | Foreign Key references Product_no of product_master table. |
| QTY_ORDERED | NUMBER(8) | |
| QTY_DISP | NUMBER(8) | |
| PRODUCT_RATE | NUMBER(10,2) | |

**Table Name :**challan_header

| Field Name | Data Type | Null |
|---|---|---|
| CHALLAN_NO | VARCHAR2(6) | Primary Key / first two letters must Start with 'CH'. |
| S_ORDER_NO | VARCHAR2(6) | Foreign Key references s_order_no of Sales_order table. |
| CHALLAN_DATE | DATE | NOT NULL |
| BILLED_YN | CHAR(1) | values('Y','N'),Default 'N'. |

**Table Name :**Challan_Details

| Field Name | Data Type | Null |
|---|---|---|
| CHALLAN_NO | VARCHAR2(6) | Foreign Key references Challan_no of challan_header table. |
| PRODUCT_NO | VARCHAR2(6) | Foreign Key references Product_no of product_master table. |
| QTY_DISP | NUMBER(4,2) | NOT NULL |

1. **. Insert following records in respective tables.**

**Table Name: CLIENT_MASTER**

| CLIENT NO. | CLIENT NAME | CITY | PINCODE | STATE | BAL_DUE |
|---|---|---|---|---|---|
| C00001 | Ivan Bayross | Bombay | 400054 | Maharashtra | 15000 |
| C00002 | Vandana Saitwal | Madras | 780001 | Tamil Nadu | 0 |
| C00003 | Pramada Jaguste | Bombay | 400057 | Maharashtra | 5000 |
| C00004 | Basu Navindgi | Bombay | 400056 | Maharashtra | 0 |
| C00005 | Ravi Shreedharan | Delhi | 100001 | | 2000 |
| C00006 | Rukmini | Bombay | 400050 | Maharashtra | 0 |

**Table Name: PRODUCT_MASTER**

| PRODUCT no | DESCRIPTION | PROFIT_PERCENT | UNIT_MEASU | QTY_ON_HAND | REORDER_LVL | SELL_PRICE | COST_PRICE |
|---|---|---|---|---|---|---|---|
| P00001 | 1.44 Floppies | 5 | Piece | 100 | 20 | 525 | 500 |
| P03453 | Monitors | 6 | Piece | 10 | 3 | 12000 | 11280 |
| P06734 | Mouse | 5 | Piece | 20 | 5 | 1050 | 1000 |
| P07865 | 1.22 Floppies | 5 | Piece | 100 | 20 | 525 | 500 |
| P07868 | Keyboards | 2 | Piece | 10 | 3 | 3150 | 3050 |
| P07885 | CD Drive | 2.5 | Piece | 10 | 3 | 5250 | 5100 |
| P07965 | 540 HDD | 4 | Piece | 10 | 3 | 8400 | 8000 |
| P07975 | 1.44 Drive | 5 | Piece | 10 | 3 | 1050 | 1000 |
| P08865 | 1.22 Drive | 5 | Piece | 2 | 3 | 1050 | 1000 |

**Table Name: SALES_MASTER**

| Salesman_No | Salesman_Name | Address1 | Address2 | City | Pincode | State | Saleamt | Tgt_To_Get | Ytd sales | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|
| S00001 | Kiran | A/14 | Worli | Bombay | 400002 | MAH | 3000 | 100 | 50 | Good |
| S00002 | Manish | 65 | Nariman | Bombay | 400001 | MAH | 3000 | 200 | 100 | Good |
| S00003 | Ravi | P-7 | Bandra | Bombay | 400032 | MAH | 3000 | 200 | 100 | Good |
| S00004 | Ashish | A/5 | Juhu | Bombay | 400044 | MAH | 3500 | 200 | 150 | Good |

**Table Name: SALES_ORDER**

| S_ORDER_NO | S_ORDER_DATE | CLIENT_NO | DELY_TYPE | BILLED | SALESMANNO | DELY_DATE | ORDER_STATUS |
|---|---|---|---|---|---|---|---|
| O19001 | 12-JAN-96 | C00001 | F | N | S00001 | 20-JAN-96 | In Process |
| O19002 | 25-JAN-96 | C00002 | P | N | S00002 | 27-JAN-96 | Canceled |
| O46865 | 18-FEB-96 | C00003 | F | Y | S00003 | 20-FEB-96 | Fulfilled |
| O19003 | 03-APR-96 | C00001 | F | Y | S00001 | 07-APR-96 | Fulfilled |
| O46866 | 20-MAY-96 | C00004 | P | N | S00002 | 22-MAY-96 | Canceled |
| O10008 | 24-MAY-96 | C00005 | F | N | S00004 | 26-MAY-96 | In Process |

**Table Name: SALES_ORDER_DETAIL**

| S_ORDER_NO | PRODUCT_NO | QTY_ORDERED | QTY_DISP | PRODUCT_RATE |
|---|---|---|---|---|
| O19001 | P00001 | 4 | 4 | 525 |
| O19001 | P07965 | 2 | 1 | 8400 |
| O19001 | P07885 | 2 | 1 | 5250 |
| O19002 | P00001 | 10 | 0 | 525 |
| O46865 | P07868 | 3 | 3 | 3150 |

| | | | | |
|---|---|---|---|---|
| O46865 | P07885 | 3 | 1 | 5250 |
| O46865 | P00001 | 10 | 10 | 525 |
| O46865 | P03453 | 4 | 4 | 1050 |
| O19003 | P03453 | 2 | 2 | 1050 |
| O19003 | P06734 | 1 | 1 | 12000 |
| O46866 | P07965 | 1 | 0 | 8400 |
| O46866 | P07975 | 1 | 0 | 1050 |
| O10008 | P00001 | 10 | 5 | 525 |
| O10008 | P07975 | 5 | 3 | 1050 |

**Table Name: challan_header**

| CHALLAN _NO | S_ORDER_NO | CHALLAN_DATE | BILLED |
|---|---|---|---|
| CH9001 | O19001 | 12-DEC-95 | Y |
| CH6865 | O46865 | 12-NOV-95 | Y |
| CH3965 | O10008 | 12-OCT-95 | Y |

**Table Name: challan_details**

| CHALLAN _NO | PRODUCT _NO | QTY_DISP |
|---|---|---|
| CH9001 | P00001 | 4 |
| CH9001 | P07965 | 1 |
| CH9001 | P07885 | 1 |
| CH6865 | P07868 | 3 |
| CH6865 | P03453 | 4 |
| CH6865 | P00001 | 10 |
| CH3965 | P00001 | 5 |
| CH3965 | P07975 | 2 |

## QUERIES

### Single Table Retrival :

| | |
|---|---|
| 1. | Find out the names of all the clients. |
| 2. | Print the entire client_master table. |
| 3. | Retrieve the list of names and the cities of all the clients. |
| 4. | List the various products available from the product_master table. |
| 5. | Find the names of all clients having ‘a’ as the second letter in their names. |
| 6. | Find out the clients who stay in a city whose second letter is ‘a’. |

| | |
|---|---|
| 7. | Find the list of all clients who stay in city 'Bombay' or city 'Delhi' or city 'Madras'. |
| 8. | List all the clients who are located in Bombay. |
| 9. | Print the list of clients whose bal_due are greater than value 10000. |
| 10. | Print the information from sales_order table of orders placed in the month of January |
| 11. | Display the order information for client_no 'C00001' and 'C00002'. |
| 12. | Find the products with description as '1.44 Drive and '1.22 Drive.' |
| 13. | Find the products whose selling price is greater than 2000 and less than or equal to 5000. |
| 14. | Find the products whose selling price is more than 1500 and also find the new selling price as original selling price*15. |
| 15. | Rename the new column in the above query as new_price. |
| 16. | Find the products whose cost_price is less than 1500. |
| 17. | List the products in sorted order of their description. |
| 18. | Calculate the square root of the price of each product. |
| 19. | Divide the cost of product '540 HDD' by difference between its price and 100. |
| 20. | List the names, city and state of clients not in the state of 'Maharashtra'. |
| 21. | List the product_no, description, sell_price of products whose description begin with letter 'M'. |
| 22. | List all the orders that were cancelled in the month of March. |
| 23. | Display all the Client names in Capitals and State in Lowercase. |
| 24. | Price must be displayed in 8,2 format. Put '*' before the amount. e.g. ***23.90. |
| 25. | Display Description and Qty_on_hand for each product. |
| 26. | Display sales transactions (qty_ordered and qty_disp) from Sales_order_details. |

## Set Functions and Concatenation:

| | |
|---|---|
| 27. | Count the total number of orders. |
| 28. | Calculate the average price of all the products. |
| 29. | Calculate the minimum price of products. |

| | |
|---|---|
| 30. | Determine the maximum and minimum product prices. Rename the title as max_price and min_price respectively. |
| 31. | Count the number of products having price greater than or equal to 1500. |
| 32. | Find all the products whose qty_on_hand is less than reorder level. |
| 33. | Print the information of client_master,product_master,sales_order table in the following format for all the records :-<br>{cust_name} has placed order {order_no} on {s_order_date} |

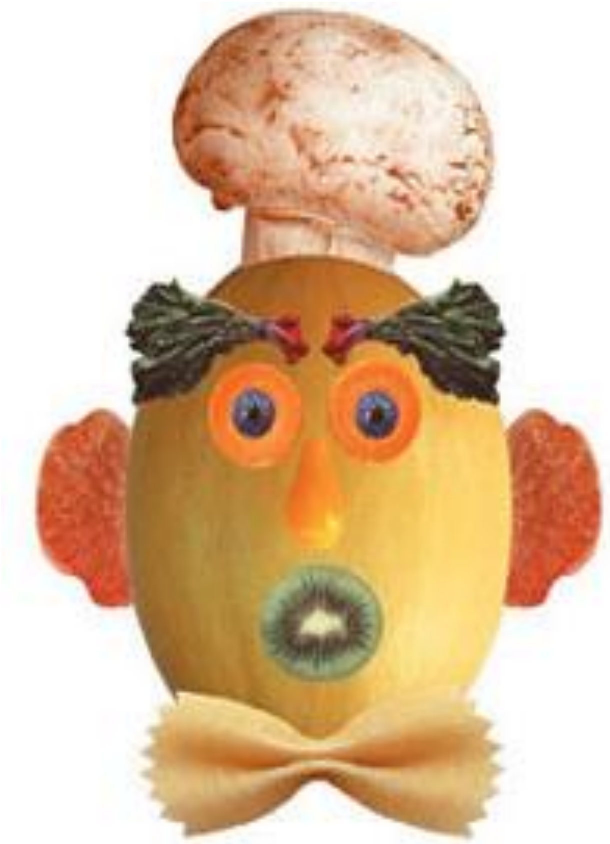
**Having and Group By:**

| | |
|---|---|
| 34. | Print the description and total qty sold for each product. |
| 35. | Find the value of each product sold. |
| 36. | Calculate the average qty sold for each client that has a maximum order value of 15000.00 |
| 37. | Find out the total sales amount receivable for the month of jan.It will be the sum total of all the billed orders for the month. |
| 38. | Print the information of product_master,order_detail table in the following format for all the records :- {description} worth Rs. {total sales for the product} was sold. |
| 39. | Print the information of product_master,order_detail table in the following format for all the records :-<br>{description} worth Rs. {total sales for the product} was ordered in the month of {s_order_date in month format} |
| 40. | Find out the total Qty_Disp and Qty_ordered of each item from Sales_order_Details. |
| 41. | Find out the no. of sales transactions of each day from Sales_order_details. |
| 42. | Compute average, minimum and maximum Qty_ordered for each product. |

# ASSIGNMENT - I

1. **Create the photo frame as given below.**

2. **Using the different veggie parts develop the following image. (Images->Q2)**



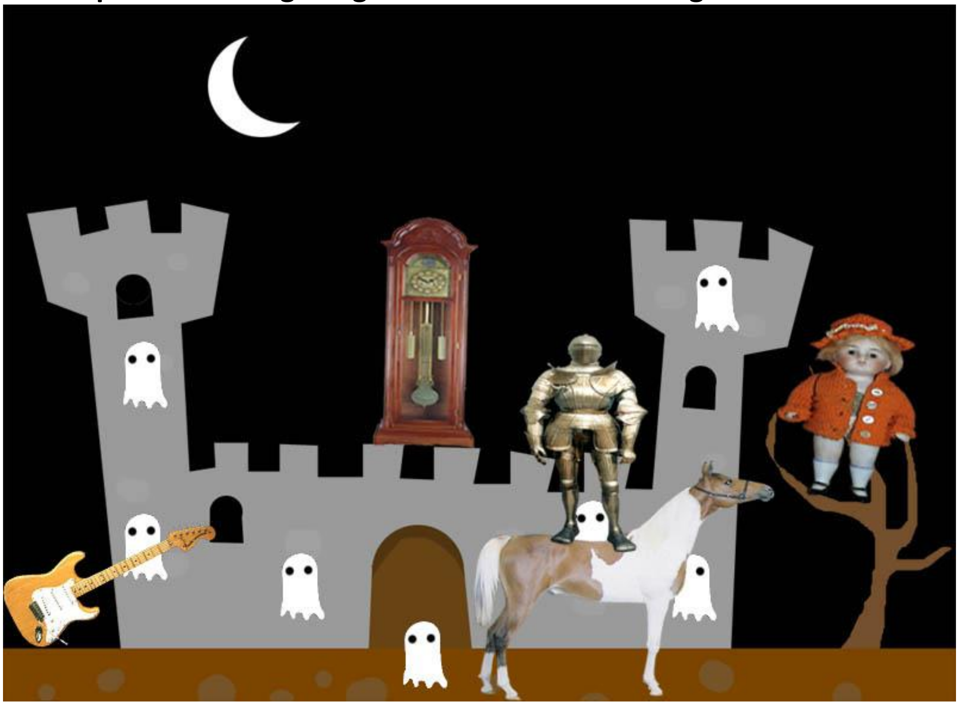
3. **sit the faces inside the frames properly.**

4. **Develop the following image from the different images.**

ssss

**5. Develop the following image from the different images.**

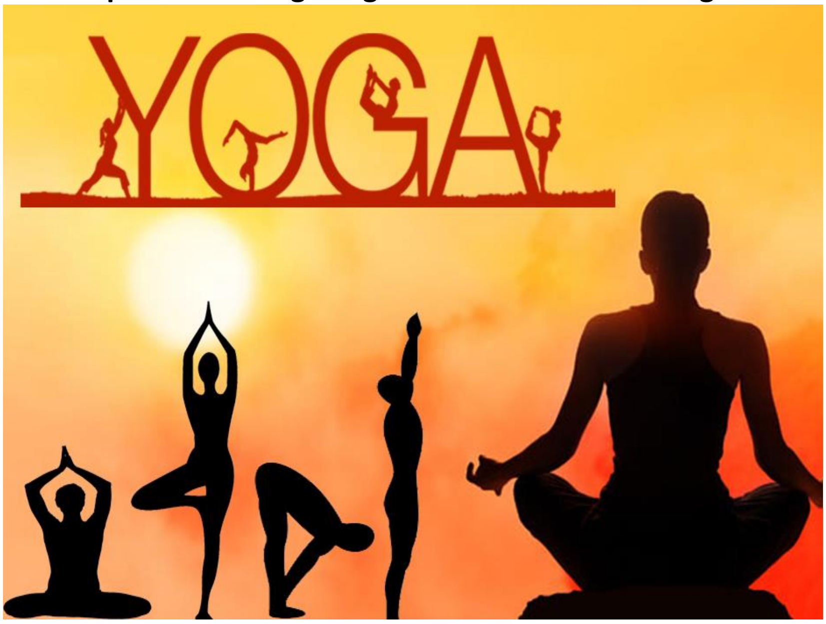
6. **Develop the following image from the different images.**



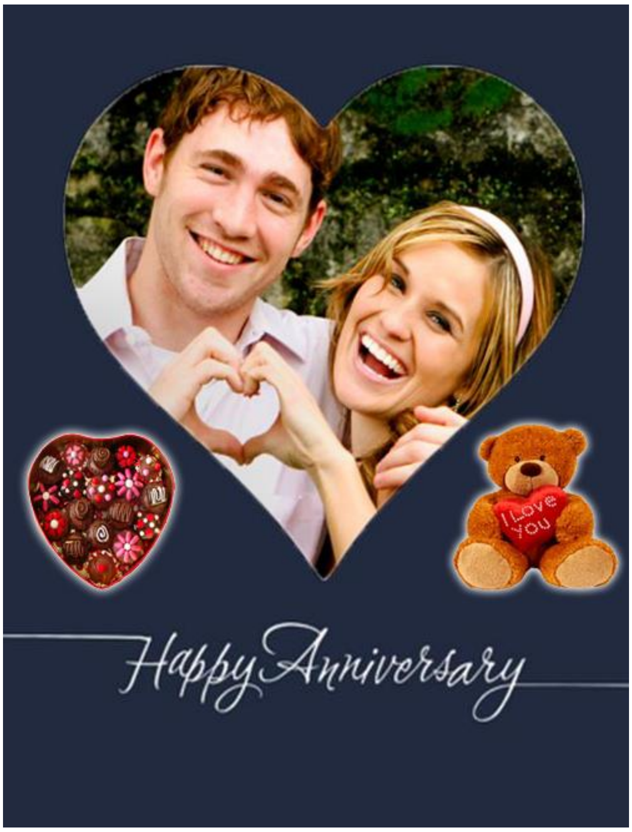
7. **Develop the following image from the different images.**

8. **Develop the following image**

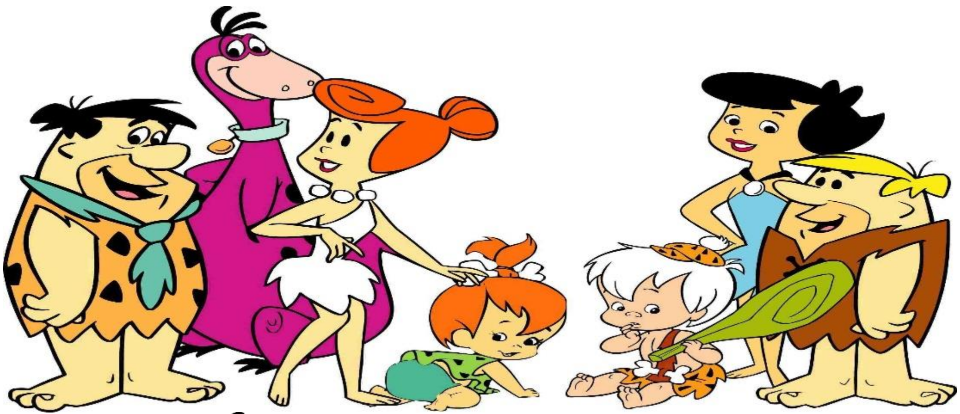

9. **Develop the following image from the different images**

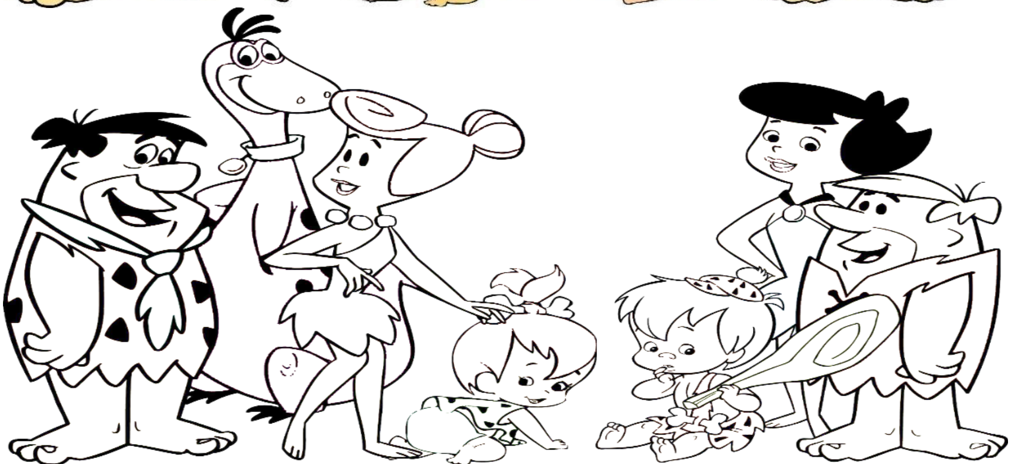



**10. Develop the following image from the different images.**


I Love You
Happy Anniversary

**11. Convert the below image1 into image2 and image3.**





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CMPICA
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## ASSIGNMENT - II

**Create this image using the different tools of Photoshop. All raw material is available in the folder which has the name Img-1, Img-2, Img-3s**


POSTER A4 TEMPLATE
Design
DO YOU NEED
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SERVICES?
DESIGN STUDIO
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OUR SERVICES
Lorem ipsum dolor sit amet, consectetur
adipiscing elit, sed do eiusmod tempor
incididunt ut labore et dolore magna aliqua.
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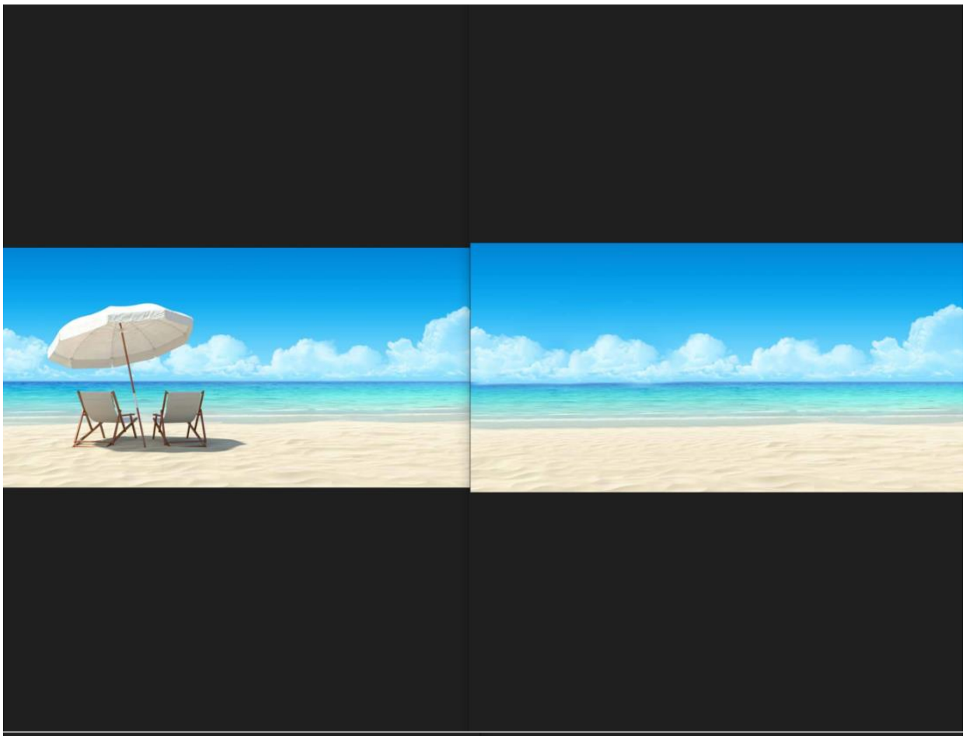
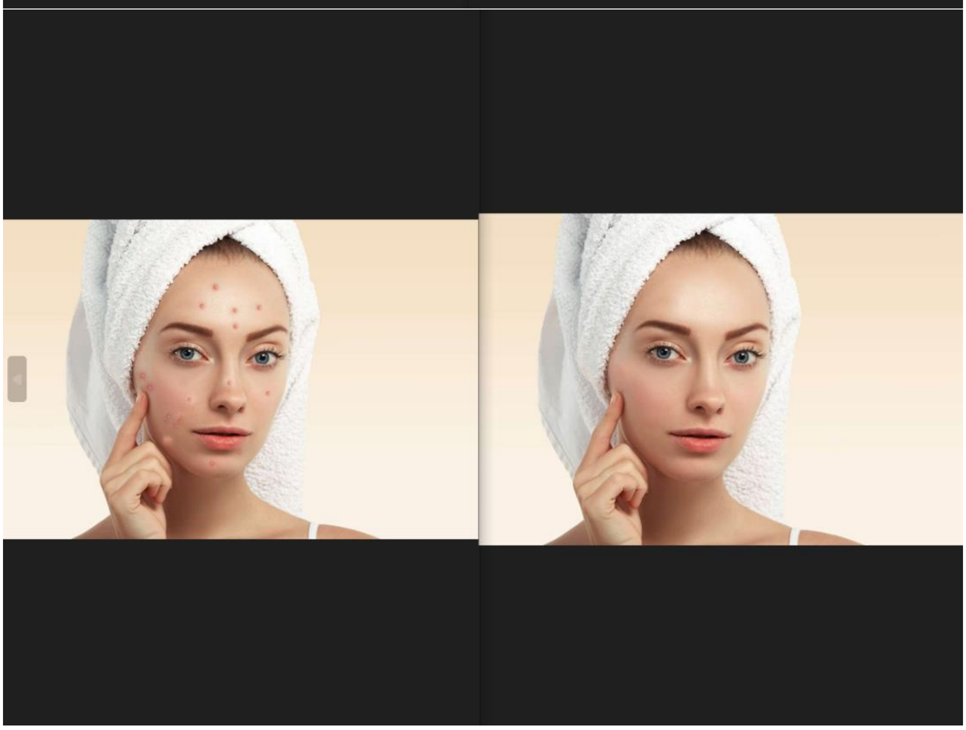
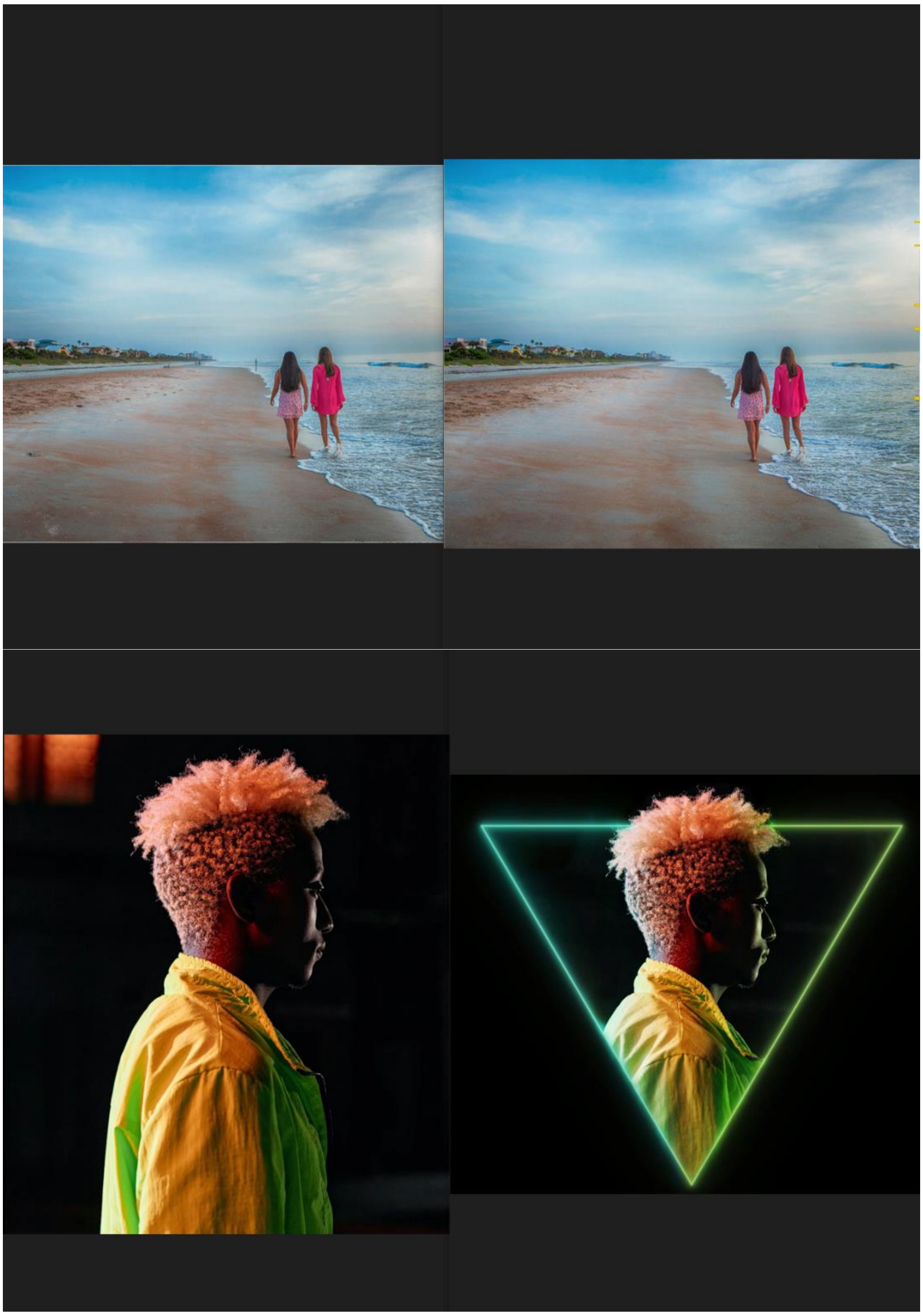
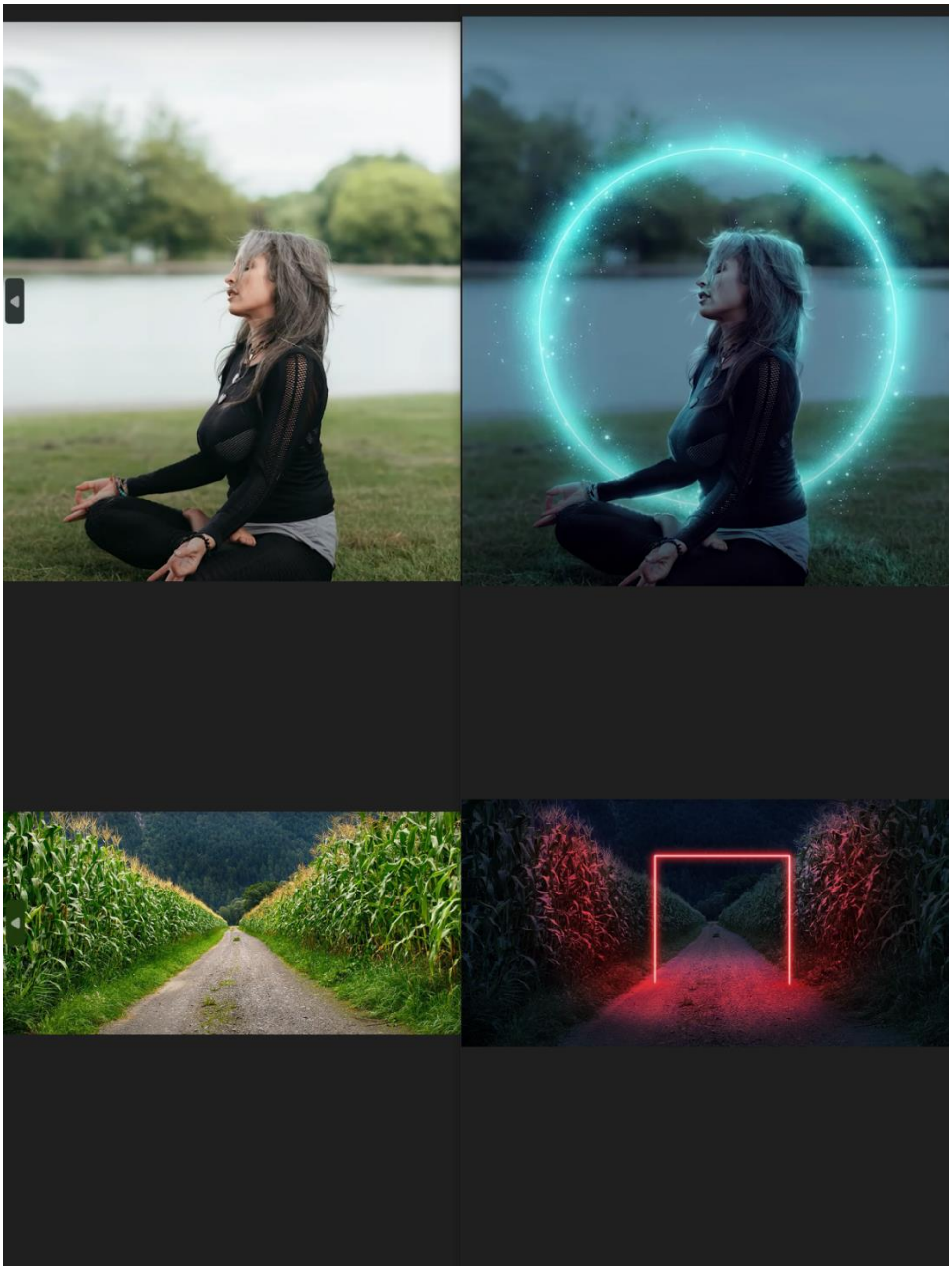
## ASSIGNMENT – III

- **Convert the left side image in to right side image using the tools available in Photoshop. All required raw material is available in folder name assignment-3**










NEON
GLOW

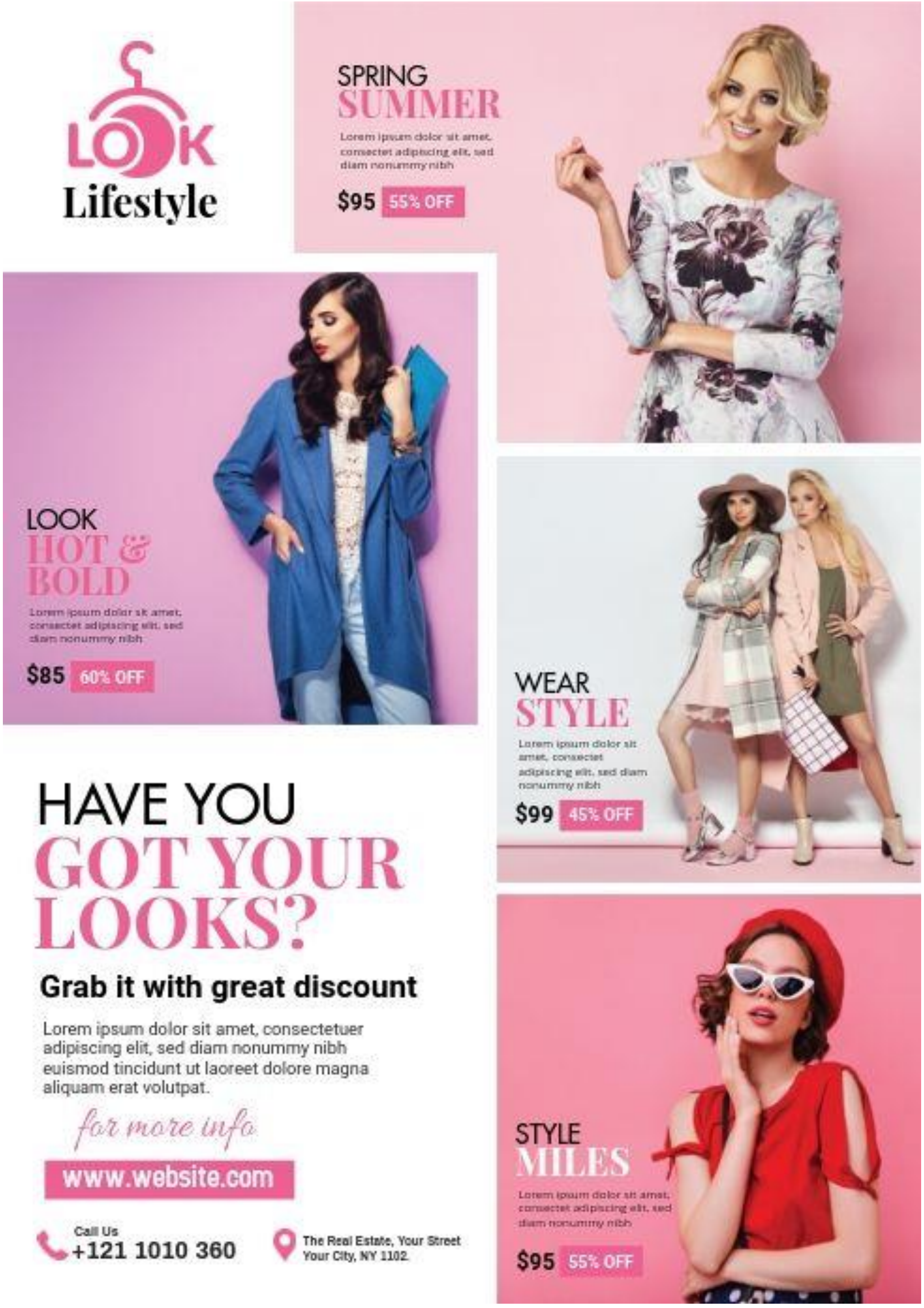
## ASSIGNMENT – V

- **Create the GIF files by using frames as per the explanation with help if the images available in folder name assignment-4.**

The picture can't be displayed.

The picture can't be displayed.


indiater.com
Suits
RAYOMDS
ARVIND
NEW
SUITS
COLLECTION
/ indiater
VISIT STORE
WWW.INDIATER.COM


LOOK
Lifestyle
SPRING
SUMMER
Lorem ipsum dolor sit amet, consectet adipiscing elit, sed diam nonummy nibh
$95 55% OFF
LOOK
HOT &
BOLD
$85 60% OFF
WEAR
STYLE
$99 45% OFF
HAVE YOU
GOT YOUR
LOOKS?
Grab it with great discount
Lorem ipsum dolor sit amet, consectetuer adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat.
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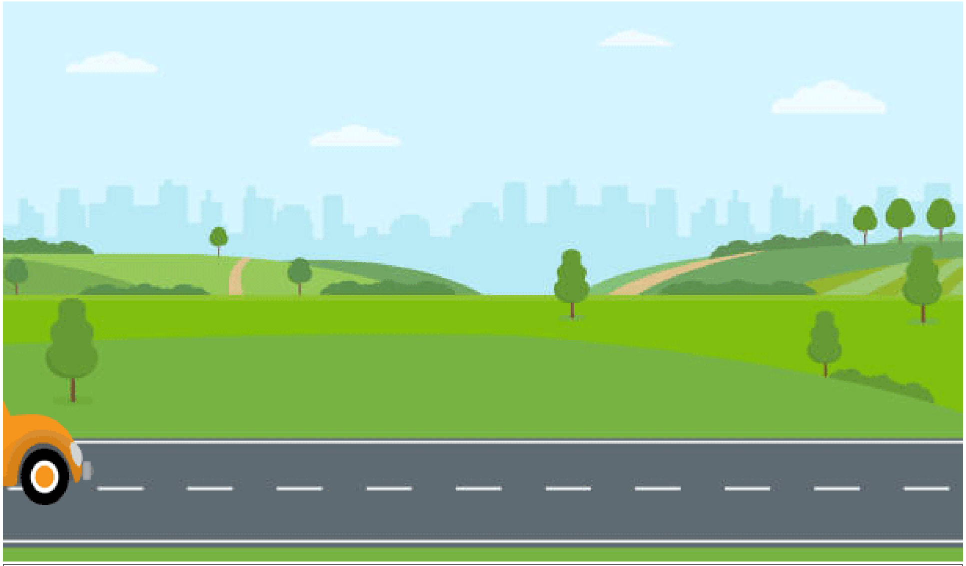
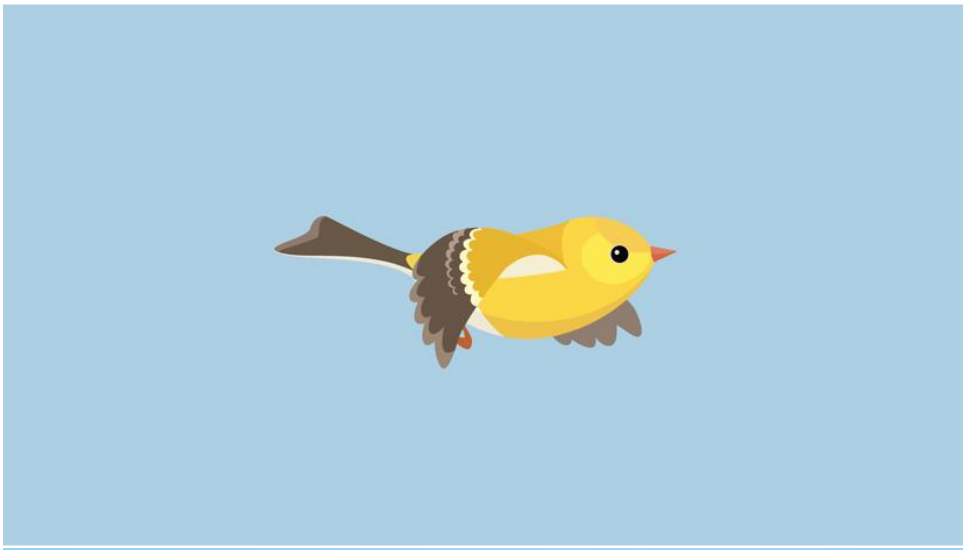
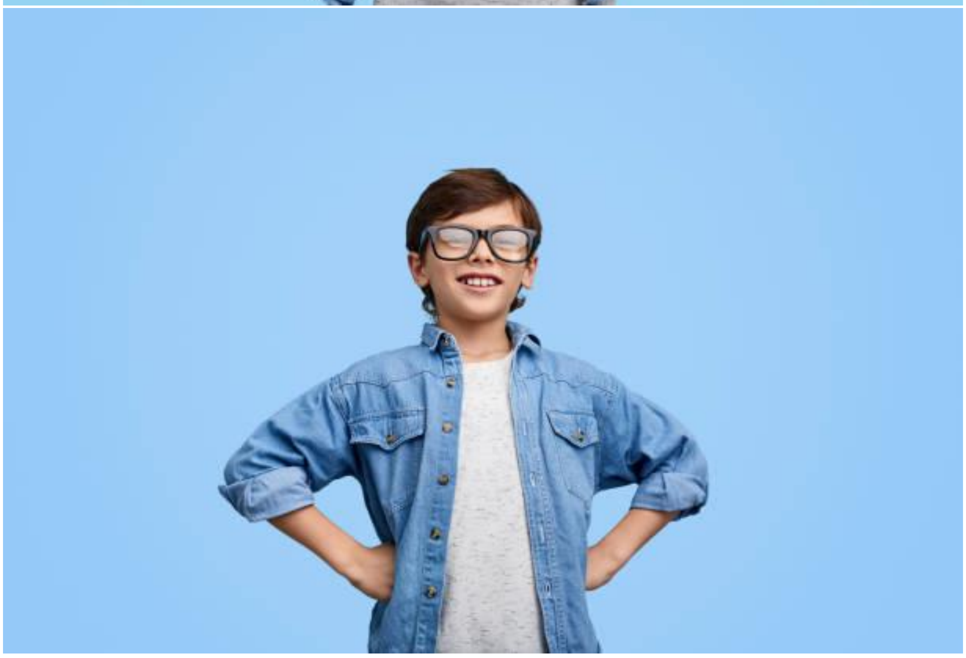
STYLE
MILES
Lorem ipsum dolor sit amet, consectet adipiscing elit, sed diam nonummy nibh
$95 55% OFF



The picture can't be displayed.






ssss

## Practical Assignment – I

## Getting Familiarity with C++

1. Write a C++ program to print Hello World.
2. Write a C++ program to display any number entered by the user.
3. Write a C++ program to input three numbers and display its total and average.
4. Write a C++ program to enter Day Temperature of 5 cities of Gujarat. Display average temperature.
5. Write a C++ program to find simple interest.
   SI = (P*R*N)/100 with float values
6. Write a C++ program to calculate area of circle.
   Area = $\pi$ * r * r
7. Write a C++ program to read two numbers from user and print their addition, subtraction, multiplication and division on screen.
8. Write a C++ program to input marks of five subjects of a student and calculate total marks and percentage.

## ASSIGNMENT -II
## Getting familiar with C++
## Class and Object

1. WAP to print your name, age and city and pin code on screen (Using Class).
2. Write a program to print the area of a rectangle by creating a class named 'Area' having two functions. First function named as 'setDim' takes the length and breadth of the rectangle as parameters and the second function named as 'getArea' returns the area of the rectangle. Length and breadth of the rectangle are entered through keyboard.

3. WAP to display addition, subtraction, multiplication and division of two integers on screen.

   Declare Class Calculation
   Declare data member num1 and num2 in Private section
   Write member function for initialize num1 and num2
   Write member function for each operation.

4. WAP to find area of circle. Area of Circle = PI * r * r Where, PI = 3.14
   (Using Class and Object)

5. Write a C++ program to create a class for student to get and print details of a student. Following are the details of a student:

   Studid, name ,sem, branch

6. Write a C++ program to demonstrate ATM money withdrawal process by taking following private data members:

   Accountno, balance;

   The withdrawal function should return remaining balance to the user and should deduct withdrawal amount from balance. If withdrawal amount > balance print appropriate message on screen (Not enough balance)

   The Deposit function should return updated balance to user.

7. Write a C++ program to create a class for student to get and print details of a student. Following are the details of a student:

   Student_id, Name, Branch,Sub1_mark,Sub2_mark,Sub3_mark,Sub4_mark,Sub5_mark

   Write member function to calculate Percentage, Class (Dist,First,Second,Pass ) of student

8. C++ program to create a class for Employee to get and display following Employee information:

   Empcode, Basicsalary

   Write member function to calculate Net salary
   DA=174% of Basic salary
   HRA=10% of Basic salary
   TA=500
   Income tax=5 % of basic salary, if basic salary >50000
   Income tax=0, if Basic salary <=50000

   Netsalary=(basicsalary+da+hra+ta) - income tax

9. Do above program for 5 number of Employees.(Using Array of Object)

10. Write a program to read time in hh:mm:ss and display answer in only seconds. For example if user enters 2:15:30 then it should display 8130 seconds.
    Input:
    Enter Hours:2
    Enter minutes:15
    Enter Seconds:30
    Output:
    8130 seconds

**ASSIGNMENT -III**
**Getting familiar with C++**
**Constructor**

1. Write a program to print the area of a rectangle by creating a class named 'Area' having one function. Length and breadth of the rectangle are entered through keyboard using Parameterized constructor.

2. WAP to display addition, subtraction, multiplication and division of two integers on screen.

   Declare Class Calculation

   Write Parameterized constructor for initialize num1 and num2

   Write member function for each operation.

3. Write a C++ program to demonstrate ATM money withdrawal and deposit process by taking following private data members:

   Accountno, balance;

   Account no and balance data member initialize using parameterized constructor

   Write three function 1. Deposit 2. Withdraw 3. Balance

   Write menu driven choice

   1. Deposit
   2. Withdraw
   3. Balance
   4. Exit

   Program stop execution when user enter choice 4.

4. Create a class “Mobile” with attributes: brand, price, color, width, height. Use constructor to set default values of these attributes. Write function to display details of all attributes.

5. Create a class “Mobile” with attributes: brand, price, color. Enter detail of five different mobile. (Using Array of object).

Display total number of mobile having price greater than 5000.

Display Brand, Price and color for all mobiles for price range 1000 to 10000

6. C++ program to create a class for Employee to get and display following Employee information:

Empcode,Emp name, Basicsalary

Count the created objects using static member.

7. Create a class "Student" having following data members:

studid, name, marks (of 5 subject), percentage.

Use getdata and show functions to input and display student data.
Create private function to calculate percentage.
Display detail of student who secured highest percentage.
Create N number of student using array of object.

8. Create a class Student with Rollno,mark1,mark2,mark3 data member and write appropriate function to setdata .Write another class name sport with sport_mark data member and set sport_mark. Find Resul(mark1+mark2+mark3+sport_mark) of student using Friend function

## ASSIGNMENT IV
## Getting familiar with Inheritance in C++
## Inheritance

1. Write a program that defines a shape class with a constructor that gives value to width and height. Then define two sub-classes triangle and rectangle, that calculate the area of the shape. In the main, define two objects a triangle and a rectangle and then call the area () function.

2. Write a program with a mother class animal. Inside it define a name and an age variables, and set_value () function. Then create two sub class Zebra and Dolphin which write a message telling the age and name of animal, also giving some extra information for both sub class (e.g. place of origin).place of origin of zebra is Earth and place of origin of dolphin is water.

3. Write a program as per following details

Create one base class Teacher

Data members Name, Department, College name, Email id.

Create sub classes for Math Teacher, English Teacher, and Science Teacher

Data member Qualification, Expertise and salary.

Display following details for each teacher

Name:

Department:

College name:

Email id:

Qualification:

Expertise:

Salary:

4. Write a program as per following details

Create one base class DRUG with following data members

Category- (i.e. **stimulants, inhalants, cannabinoids)**

Date_of_manufacture, Company name

Create one sub class TABLET derived from DRUG with following data members

Tablet name, Price

Create one sub class PainReliever derived from TABLET with data member

Dosage_units: i.e( 1 or 2 or 3)
Side_effects : i.e (Nausea, Drowsiness, Dizziness.)

Use_within_days: i.e(10 or 20 or 30).

Use appropriate member function for setting and Getting above details and display details in main function.

5. Write a program as per following details

Create one base class HOTEL with following data members

Hotel_name,

Hotel_type i.e(Three star,five star)

City

Hotel_rate i.e(2000,3000,5000)

Create one base class FLIGHT with following data members

Flight_no

Source city

Destination city

Seat no

Create one sub class PASSENGER derived from HOTEL and FLIGHT with following data members

Name, Age, city

Write appropriate member functions in each class and display all information in main

6. Write a program as per following details

Create one base class PERSON with following data members

Name, College name

Create one sub class STUDENT derived from PERSON with following data members

Student_id , Marks of five subject, percentage

Member function:

showResult( )-Calculate total,percentage and finding class(Dist,First,second,pass)

Create one sub class EMPLOYEE derived from PERSON with following data members

Emp_id, qualification , basic salary

Member function to calculate Net salary and print Net salary

DA=189% of Basic salary
HRA=10% of Basic salary
TA=500
Income tax=5 % of basic salary, if basic salary >50000
Income tax=0, if Basic salary <=50000

Netsalary=(basicsalary+da+hra+ta) - income tax

Write appropriate setter function in each class and display detail of student and employee in main.

**ASSIGNMENT V**
**Getting familiar with Operator Overloading**

1. Write a program to overload unary minus operator

Ex. Input : obj(10,-20,30) Output : -10,20,-30 (-obj)

2. Write a program to overload pre decrement operator (obj1= --obj)

3. Write a program to overload post decrement operator (obj1=obj--)

4. Write a program to overload Binary *(multiply) operator for object of same class

Ex. class student s1,s2,s3 s3=s1*s2

5. Write a program to overload Binary * (multiply) operator for object of different class

Ex. Class unit_test, class Practical Practical result=object of unit_test+object of practical

6. Write program to overload = = operator to compare two object of student class

```
Class student
{
        Int mark;
        Public:
       Student (int tmark)
      {
        mark=tmark;
      }
};

Int main() {

Student s1(100),student s2(200),student s3(100);

If(s1==s2)

   Cout<<" no same marks"

If(s1==s3)

  Cout<< "same marks"
```

# ASSIGNMENT VI
# Working with File

1.Write a program to create a file “Product.txt” and write data as Product Name, Product Price and Batch Number in a file.

2. Write a program to append Product Type and Product Weight in a file “Product.txt”.

3. Write a program to create a file “Patient.txt” and enter Patient ID, Patient Name, Disease Name and City in a file. Now read patient details and display all the details on a terminal.

4. Write a program to append Doctor Name and Doctor Speciality in “Patient.txt” file and display the data on a console.

5.Create a class “Mobile” with attributes: brand, price, color. Use getData() values of these attributes and store object in file. Write function displayData() to display object from file.

6. Create a class “Student” having following data members:

studid,name,marks (of N subject), percentage. Create private function to calculate percentage. Use getData() functions get data from keyboard and store data to the file. Write function displayData() to display object from file.